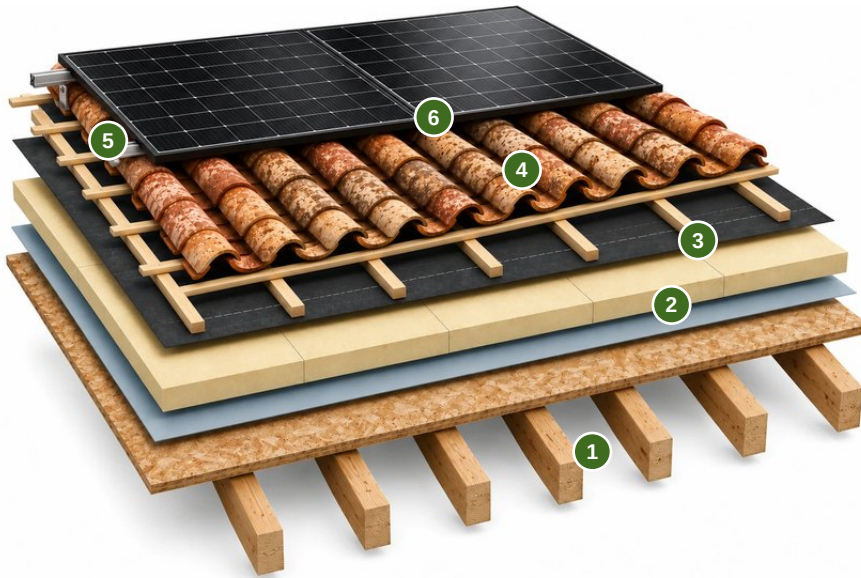




SOLAR ON A TRADITIONAL TILED ROOF

A layered roof, a precise connection and a discreet array. Solar is installed above the tiles while maintaining the roof's drainage and ventilation.



SYSTEM OVERVIEW

The roof remains the weatherproof envelope. Above it, the mounting system supports the modules and transfers loads to the structure. Electrical conversion, protection and optional storage complete the installation.

Illustrative integration

ROOF BUILD-UP & SOLAR INTEGRATION

- | | |
|---------------------------------|---------------------------------|
| 1 Structural timber & roof deck | 4 Terracotta roof tiles |
| 2 Vapour control & insulation | 5 Structural roof hooks & rails |
| 3 Underlay & ventilated battens | 6 Photovoltaic modules |

TECHNICAL PRINCIPLES

Design the support system for the roof structure and wind exposure. Preserve waterproofing, drainage and maintenance access. Confirm equipment, protections, metering and registration with the appointed electrical specialist.

Example insulated roof build-ups. Existing roofs may have a different construction; solar installation does not automatically include new insulation or a roof renovation.

Roof & structure

Tile condition, supporting structure, fixing points, roof load, membrane compatibility, drainage and maintenance access.

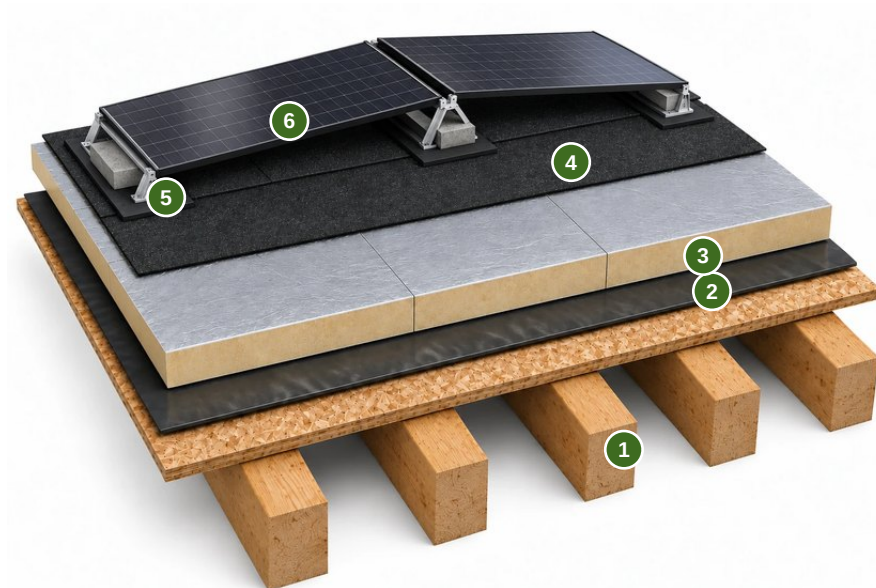
Wind & exposure

Layout, roof edge zones, attachment or ballast calculations and corrosion resistance for the site.



SOLAR ABOVE AN INSULATED FLAT ROOF

Low-profile supports sit above the waterproofing. Protection pads, roof capacity, drainage and wind design determine the mounting and ballast.



Illustrative integration

SYSTEM OVERVIEW

The roof remains the weatherproof envelope. Above it, the mounting system supports the modules and transfers loads to the structure. Electrical conversion, protection and optional storage complete the installation.

ROOF BUILD-UP & SOLAR INTEGRATION

- | | |
|---------------------------------|------------------------------------|
| 1 Structural joists & roof deck | 4 Continuous waterproofing |
| 2 Vapour control layer | 5 Protection pads & support system |
| 3 Rigid thermal insulation | 6 Photovoltaic modules |

TECHNICAL PRINCIPLES

Design the support system for the roof structure and wind exposure. Preserve waterproofing, drainage and maintenance access. Confirm equipment, protections, metering and registration with the appointed electrical specialist.

Example insulated roof build-ups. Existing roofs may have a different construction; solar installation does not automatically include new insulation or a roof renovation.

Roof & structure

Tile condition, supporting structure, fixing points, roof load, membrane compatibility, drainage and maintenance access.

Wind & exposure

Layout, roof edge zones, attachment or ballast calculations and corrosion resistance for the site.



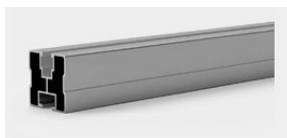
THE DETAILS THAT MAKE THE INSTALLATION.

Original ProMount product images from the TLS-Solutions material proposal. Final dimensions, fixing spacing, ballast and article references are specified for the installation.



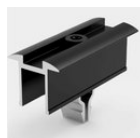
1. Double-adjustable roof hook

ProMount adjustable connection for pitched roofs.



2. Aluminium mounting rail

ProMount rail supporting the panel array.



3. Click middle clamp

ProMount clamp between adjacent modules.



4. Click end clamp

ProMount fixing at the end of the panel row.



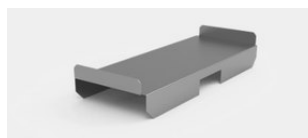
5. SOLARSPEED east-west profile

ProMount support profile for opposing module orientations.



6. SOLARSPEED side plate profile

Matching side plate for the selected support layout.



7. Ballast holder

ProMount tray for the calculated ballast.

SELECTED EQUIPMENT. A CLEARLY DEFINED INSTALLATION.



Bauer glass-glass panels

465 Wp modules form the basis of our example packages. Bauer publishes a 30-year product and performance warranty, subject to the applicable model and conditions.



Solis inverter & monitoring

The inverter is matched to the supply and panel layout. Monitoring is specified with the appropriate communication equipment and internet connection.



Marstek energy storage

An optional AC-coupled storage route. Storage capacity, usable power and backup arrangements are specified for the selected model.

WHAT IS CHECKED BEFORE INSTALLATION?

Design the support system for the roof structure and wind exposure. Preserve waterproofing, drainage and maintenance access. Confirm equipment, protections, metering and registration with the appointed electrical specialist.

Electrical design

Supply phases, inverter compatibility, string design, cable routes, protective devices, earthing and measurement.

Documentation & handover

Equipment references, design scope, checks, installation certificate, registration, monitoring access and warranty documents as agreed.



THREE SETTINGS. ONE CAREFULLY DESIGNED SYSTEM.



Pitched roofs

ProMount rails, hooks and clamps selected for the tiles and supporting structure, with careful roof connections.



Flat roofs

ProMount SOLARSPEED layouts, including east-west options. Roof capacity, wind loads, ballast, drainage and membrane protection determine the design.



Ground-mounted systems

A property-specific option where roof installation is unsuitable. Structure, foundations, cable route and planning requirements are assessed separately.

FROM SUNLIGHT TO USEFUL ELECTRICITY

1. Modules & DC circuit

Bauer 465 Wp glass-glass modules. Layout, string voltage and current must suit the selected inverter. Connector compatibility and cable protection are checked.

2. Inverter

A suitable Solis inverter converts DC generation to AC. The exact model, phase configuration, power and communication accessories are confirmed for the installation.

3. Protection, metering & distribution

DC and AC protection, earthing, cable sizes and metering follow the installation design. Connection and certification are performed by the appointed authorised specialist.

4. Household consumption

Electricity is used by the property while it is produced. Useful system size depends on the consumption profile and occupancy.

5. Grid interface

Import, export and surplus compensation depend on the selected self-consumption arrangement, registration and electricity contract.

6. Optional AC-coupled storage

Marstek VENUS E is an AC-coupled storage option. Version, controls and metering are confirmed; whole-house backup is not assumed.

WHAT IS CHECKED BEFORE INSTALLATION?

Roof & structure

Tile condition, supporting structure, fixing points, roof load, membrane compatibility, drainage and maintenance access.

Wind & exposure

Layout, roof edge zones, attachment or ballast calculations and corrosion resistance for the site.

Electrical design

Supply phases, inverter compatibility, string design, cable routes, protective devices, earthing and measurement.

Documentation & handover

Equipment references, design scope, checks, installation certificate, registration, monitoring access and warranty documents as agreed.

Illustrative installation concepts; not completed EcoViva projects.